CP2.7 — Methods in Differential Equations

1. First-Order Linear ODEs — Integrating Factor Method

A first-order linear ODE has the form:

$$\frac{dy}{dx} + P(x)\,y = Q(x)$$

The **integrating factor** is $\mu(x) = e^{\int P(x)\,dx}$.
Multiplying both sides by μ: $\frac{d}{dx}(\mu y) = \mu Q(x)$.
Integrate both sides to find y.

1.1 Method Summary

1. Write the ODE in standard form: $\frac{dy}{dx} + P(x)y = Q(x)$
2. Compute $\mu = e^{\int P\,dx}$ (ignore the constant of integration here)
3. Multiply both sides by μ: $\frac{d}{dx}(\mu y) = \mu Q$
4. Integrate both sides: $\mu y = \int \mu Q\,dx + C$
5. Solve for y

Example: Solve $\frac{dy}{dx} + \frac{2y}{x} = x^3$.

$P(x) = 2/x$, so $\mu = e^{\int 2/x\,dx} = e^{2\ln x} = x^2$.
Multiply: $\frac{d}{dx}(x^2 y) = x^5$.

Integrate: $x^2 y = \frac{x^6}{6} + C$.

Solution: $y = \frac{x^4}{6} + Cx^{-2}$.

Example: Solve $\frac{dy}{dx} - 3y = e^{4x}$ with $y(0) = 1$.

$P = -3$, $\mu = e^{-3x}$.
$\frac{d}{dx}(e^{-3x}y) = e^x$.
$e^{-3x}y = e^x + C$, so $y = e^{4x} + Ce^{3x}$.
Apply $y(0) = 1$: $1 = 1 + C \Rightarrow C = 0$. Solution: $y = e^{4x}$.

Exam tip: Only omit the constant when computing μ. Always reintroduce $+C$ when integrating the right-hand side.

2. Second-Order Homogeneous ODEs

The general form is $a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = 0$.
Try $y = e^{rx}$. The **auxiliary (characteristic) equation** is:

$$ar^2 + br + c = 0$$

2.1 Three Cases

Discriminant $b^2 - 4ac$	Roots	General Solution
> 0 (two distinct real roots α, β)	$r = \alpha, \beta$	$y = Ae^{\alpha x} + Be^{\beta x}$
$= 0$ (repeated real root α)	$r = \alpha$ (twice)	$y = (A + Bx)e^{\alpha x}$
< 0 (complex roots $\alpha \pm \beta i$)	$r = \alpha \pm \beta i$	$y = e^{\alpha x}(A\cos\beta x + B\sin\beta x)$

Case 1 — Distinct real roots: Solve $y'' - 5y' + 6y = 0$.

AE: $r^2 - 5r + 6 = 0 \Rightarrow (r-2)(r-3) = 0 \Rightarrow r = 2, 3$.
General solution: $y = Ae^{2x} + Be^{3x}$.

Case 2 — Repeated root: Solve $y'' - 4y' + 4y = 0$.

AE: $r^2 - 4r + 4 = 0 \Rightarrow (r-2)^2 = 0 \Rightarrow r = 2$ (repeated).
General solution: $y = (A + Bx)e^{2x}$.

Case 3 — Complex roots: Solve $y'' - 4y' + 13y = 0$.

AE: $r^2 - 4r + 13 = 0 \Rightarrow r = \dfrac{4 \pm \sqrt{16-52}}{2} = \dfrac{4 \pm \sqrt{-36}}{2} = 2 \pm 3i$.
$\alpha = 2, \beta = 3$.
General solution: $y = e^{2x}(A\cos 3x + B\sin 3x)$.

3. Second-Order Non-Homogeneous ODEs

$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = f(x)$$

General solution = Complementary function (CF) + Particular integral (PI)
The CF is the general solution of the homogeneous equation (RHS = 0).
The PI is any specific solution of the full equation.

3.1 Finding Particular Integrals — Trial Function Table

Form of $f(x)$	Trial PI	Notes
k (constant)	$y_p = a$	Constant

$kx + m$ (linear)	$y_p = ax + b$	Linear polynomial
$kx^2 + mx + n$	$y_p = ax^2 + bx + c$	Match degree
ke^{px}	$y_p = ae^{px}$	Unless p is a root of AE
$k\cos px$ or $k\sin px$	$y_p = a\cos px + b\sin px$	Include both sin and cos
$ke^{px}\cos qx$	$y_p = e^{px}(a\cos qx + b\sin qx)$	Modulated oscillation

Resonance (failure) case: If the trial PI duplicates a term in the CF, multiply the trial by x (or x^2 if still a duplicate).

3.2 Worked Examples

Example — polynomial RHS: Solve $y'' - 3y' + 2y = 4x$.

CF: AE $r^2 - 3r + 2 = 0 \Rightarrow (r-1)(r-2) = 0$. CF: $y_c = Ae^x + Be^{2x}$.
PI: Try $y_p = ax + b$. Then $y_p'' = 0$, $y_p' = a$.
Substitute: $0 - 3a + 2(ax + b) = 4x \Rightarrow 2ax + (2b - 3a) = 4x + 0$.
Equate: $2a = 4 \Rightarrow a = 2$, and $2b - 3a = 0 \Rightarrow b = 3$.
PI: $y_p = 2x + 3$.
General solution: $y = Ae^x + Be^{2x} + 2x + 3$.

Example — exponential RHS: Solve $y'' + y' - 2y = 3e^{3x}$.

CF: AE $r^2 + r - 2 = 0 \Rightarrow (r+2)(r-1) = 0$. CF: $y_c = Ae^{-2x} + Be^x$.
PI: Try $y_p = ae^{3x}$. Then $y_p'' = 9ae^{3x}$, $y_p' = 3ae^{3x}$.
Substitute: $9ae^{3x} + 3ae^{3x} - 2ae^{3x} = 3e^{3x} \Rightarrow 10a = 3 \Rightarrow a = 3/10$.
General solution: $y = Ae^{-2x} + Be^x + \dfrac{3}{10}e^{3x}$.

Example — trigonometric RHS: Solve $y'' + 4y = \sin 2x$.

CF: AE $r^2 + 4 = 0 \Rightarrow r = \pm 2i$. CF: $y_c = A\cos 2x + B\sin 2x$.
Resonance! Trial PI $a\cos 2x + b\sin 2x$ is in the CF.
Try $y_p = x(a\cos 2x + b\sin 2x)$.
$y_p' = a\cos 2x + b\sin 2x + x(-2a\sin 2x + 2b\cos 2x)$
$y_p'' = -4a\sin 2x + 4b\cos 2x + x(-4a\cos 2x - 4b\sin 2x)$
$y_p'' + 4y_p = (-4a\sin 2x + 4b\cos 2x) = \sin 2x$
Compare: $-4a = 1 \Rightarrow a = -1/4$; $4b = 0 \Rightarrow b = 0$.
PI: $y_p = -\dfrac{x\cos 2x}{4}$.
General solution: $y = A\cos 2x + B\sin 2x - \dfrac{x\cos 2x}{4}$.

Example — repeated root resonance: Solve $y'' - 4y' + 4y = e^{2x}$.

CF: AE $(r-2)^2 = 0$. CF: $y_c = (A + Bx)e^{2x}$.
Trial ae^{2x} duplicates CF. Trial axe^{2x} also duplicates CF.
Try $y_p = ax^2e^{2x}$.
$y_p' = (2ax + 2ax^2)e^{2x}$, $y_p'' = (2a + 8ax + 4ax^2)e^{2x}$.
Substitute: $(2a + 8ax + 4ax^2)e^{2x} - 4(2ax + 2ax^2)e^{2x} + 4ax^2e^{2x} = e^{2x}$
$2ae^{2x} = e^{2x} \Rightarrow a = 1/2$.
General solution: $y = (A + Bx)e^{2x} + \dfrac{x^2}{2}e^{2x}$.

Exam tip: Resonance occurs when the trial PI is already a solution of the homogeneous equation. Multiply by x once (or x^2 for double resonance) to get a linearly independent PI.

4. Applying Initial and Boundary Conditions

After finding the general solution $y = y_c + y_p$, apply conditions to find the arbitrary constants A and B.

Example: Solve $y'' - 5y' + 6y = 2$ with $y(0) = 1$, $y'(0) = 0$.

CF: $Ae^{2x} + Be^{3x}$. PI: try $y_p = a$: $6a = 2 \Rightarrow a = 1/3$. GS: $y = Ae^{2x} + Be^{3x} + \frac{1}{3}$.
$y(0) = 1$: $A + B + \frac{1}{3} = 1 \Rightarrow A + B = \frac{2}{3}$. \quad (i)

$y' = 2Ae^{2x} + 3Be^{3x}$. $y'(0) = 0$: $2A + 3B = 0$. \quad (ii)
From (i): $A = \frac{2}{3} - B$. Sub into (ii): $\frac{4}{3} - 2B + 3B = 0 \Rightarrow B = -\frac{4}{3}$, $A = 2$.
Solution: $y = 2e^{2x} - \frac{4}{3}e^{3x} + \frac{1}{3}$.

5. Summary — Complete Method

Step	Action
1	Write homogeneous equation and find auxiliary equation
2	Solve AE to classify roots, write CF
3	Choose trial PI based on form of $f(x)$ — check for resonance
4	Differentiate PI, substitute, equate coefficients
5	GS = CF + PI
6	Apply ICs/BCs to find A, B

6. Exam-Style Practice Questions

1. **[1st order IF]** Solve $x\dfrac{dy}{dx} + 2y = x^3 \ln x$, $x > 0$, given $y(1) = 0$.
2. **[2nd order homogeneous]** Find the general solution of $2y'' + 5y' + 2y = 0$. Describe the behaviour as $x \to \infty$.
3. **[Complex roots]** Solve $y'' + 2y' + 5y = 0$ with $y(0) = 1$, $y'(0) = 0$.
4. **[Non-homogeneous poly]** Find the general solution of $y'' - 2y' - 3y = x^2 + 1$.
5. **[Non-homogeneous exp]** Solve $y'' - 3y' + 2y = e^x$ (resonance case). Apply $y(0) = 0$, $y'(0) = 1$.
6. **[Non-homogeneous trig]** Find the general solution of $y'' + 9y = 4\cos 3x$ (resonance).
7. **[Mixed]** Solve $y'' + 4y' + 4y = 8e^{-2x}$. Apply $y(0) = 0$, $y'(0) = 4$.

General exam tips for CP2.7:

- Always find the CF first — then address the PI
- For trig PI: always use both $\cos$ AND $\sin$ in the trial, even if only one appears in $f(x)$
- Apply ICs after finding the full GS, not after finding the CF alone
- Show all differentiation steps for the PI — examiners give method marks for substitution
- Check: substituting your GS back into the ODE should give $f(x)$ identically